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CBSE CLASS 10 | MATHEMATICS

CIRCLES

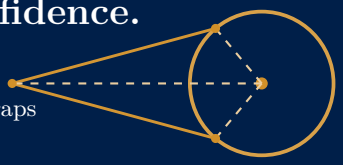
Master Concept, Formula & Visual Problem-Solving Handbook

Academic Year 2026–27

See the point. Draw the tangent. Prove with confidence.
From intuitive geometry to rigorous board proofs.

Theorems 10.1 & 10.2 | Vector TikZ Diagrams | Standard Board Proofs | Exam Traps

A student-friendly Maths Manthan revision resource



What this handbook includes

- Complete definitions: Secant vs. Tangent & Point of Contact.
- Rigorous proofs of **Theorem 10.1** (Radius ⊥ Tangent) and **Theorem 10.2** (Equal Tangents).
- Right-triangle Pythagorean models & angle relationships.
- Circumscribed polygon master results ($AB + CD = BC + DA$).
- Concentric circle chords and incircle radius formulas.

How to use this handbook

Always mark the radius to the point of contact first to expose the 90° angle. Identify whether the problem relies on **Pythagoras** (length calculations) or **cyclic properties / isosceles triangles** (angle determinations). Write full geometric reasons in board exam answers.

Official CBSE Curriculum Scope (Academic Year 2026–27)

Zero Deleted Content in Chapter 10: The rationalized NCERT Class 10 curriculum retains Chapter 10: *Circles* in its entirety under Unit IV: Geometry. Both core theorems and their applications are strictly tested. Note: Geometric constructions of tangents (formerly Chapter 11) remain **permanently excluded** from the syllabus. Chord theorems and cyclic quadrilaterals belong to Class 9 and serve solely as background tools. Source: CBSE Secondary Curriculum (2026–27), Unit IV: Geometry.

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Prepared for learning and revision – not an official CBSE publication.

CHAPTER 10

SUPPORTING SKILL

FOUNDATIONAL GEOMETRY OF CIRCLES

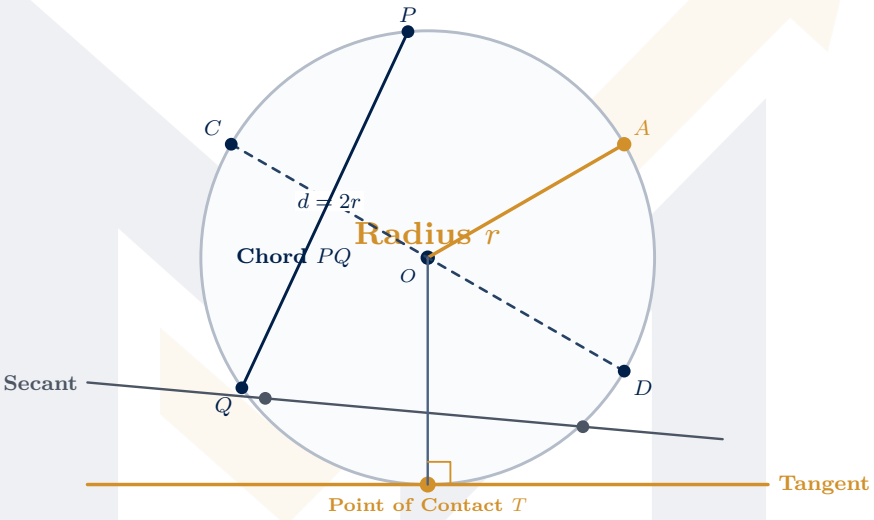
Radius, Diameter, Chord, Secant, and Tangent – The Vocabulary of Circles

Module 1.1 | The Circle as a Geometric Locus

A **circle** is the collection of all points in a plane equidistant from a fixed point (centre O).

- **Centre (O):** Fixed point. **Radius (r):** Fixed distance from centre to boundary ($r > 0$).
- **Three Regions:** Interior ($OP < r$), On the circle ($OP = r$), and Exterior ($OP > r$).

Module 1.2 | Anatomy of Line Segments Associated with a Circle



Module 1.3 | Line-Circle Relationship Taxonomy

Let d be the perpendicular distance from centre O to line l:

Relationship	Condition	Common Points	Geometric Identity and Property
Non-Intersecting	$d > r$	0	Line lies completely outside the circle. No intersection.
Secant	$d < r$	2	Line cuts the circle at two distinct points. Chord is internal segment.
Tangent	$d = r$	1	Line touches circle at **one unique point**. Limiting case of secant.

Pedagogical Insight: Tangent as the Limiting Position of a Secant

Imagine secant AB cutting the circle at A and B. As secant line rotates such that chord length $AB \to 0$, points A and B merge into a single coincident point. The secant smoothly transitions into a **tangent**. This provides the foundational intuition of tangent lines in calculus and geometry.

CHAPTER 10

CORE SYLLABUS

THE TANGENT TO A CIRCLE

Definition, Point of Contact & The Three Positional Cases

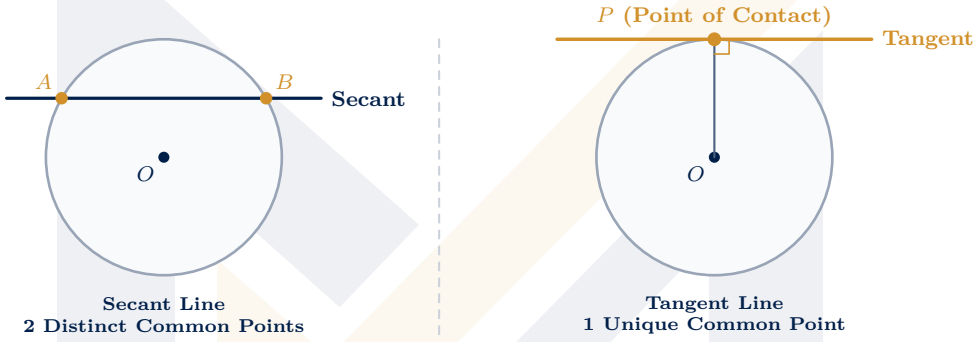
Module 2.1 | Formal Definition of a Tangent

A **tangent** to a circle is a straight line that touches the circle at **only one point**:

$$l \cap C(O,r) = \{P\}$$

The unique point P common to the line and the circle is called the **Point of Contact** (or *Point of Tangency*).

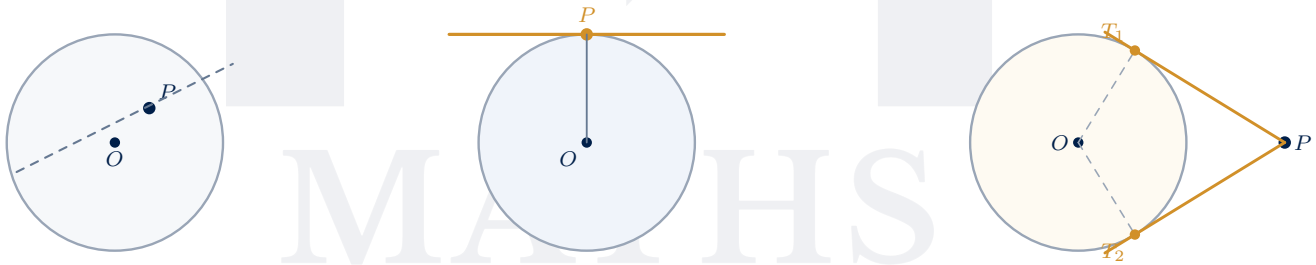
Module 2.2 | Geometric Comparison: Secant vs. Tangent



Secant Line
2 Distinct Common Points

Tangent Line
1 Unique Common Point

Module 2.3 | Number of Tangents from a Point to a Circle (The 3 Positional Cases)



Case 1: P Inside
$OP < r$
ZERO Tangents

Case 2: P On Circle
$OP = r$
EXACTLY ONE Tangent

Case 3: P Outside
$OP > r$
EXACTLY TWO Tangents

Board Exam Objective Trap: Tangent Counts

- "Tangents through a point **lying on** the circle?" $\longrightarrow$ **1**.
- "Tangents **from an external point**?" $\longrightarrow$ **2**.
- "Maximum number of parallel tangents to a circle?" $\longrightarrow$ **2** (at opposite ends of a diameter).
- "Total number of tangents that can be drawn to a circle?" $\longrightarrow$ **Infinitely many**.

CHAPTER 10

CORE SYLLABUS

THEOREM 10.1: RADIUS ⊥ TANGENT

The Fundamental Orthogonality Property at the Point of Contact

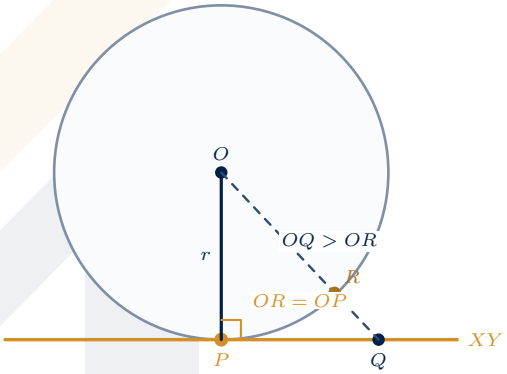
Module 3.1 | Official Theorem Statement

CBSE Theorem 10.1:
The tangent at any point of a circle is perpendicular to the radius through the point of contact.

$$\boxed{OP \perp XY \quad \Longleftrightarrow \quad \angle OPX = \angle OPY = 90^\circ}$$

where O is the centre, P is the point of contact, and XY is the tangent line.

Module 3.2 | Standard CBSE Board Proof (Shortest Distance Property)



Given: A circle $C(O, r)$ and a tangent XY touching at P.
To Prove: $OP \perp XY$.
Construction: Take any point Q on XY ($Q \neq P$). Join OQ, intersecting the circle at point R.

Proof Steps:

1. Point Q must lie **outside** the circle. (If Q were inside, XY would be a secant, not a tangent).
2. Therefore, $OQ > OR$.
3. But $OR = OP$ (radii of the same circle).
4. Hence, $OQ > OP$ for every point $Q \in XY$ ($Q \neq P$).
5. Thus, OP is the **shortest distance** from centre O to any point on line XY.
6. Since the shortest distance from a point to a line is the perpendicular distance:

OP ⊥ XY. [Q.E.D.]

Module 3.3 | Vital Converse and Diameter Corollary

Converse of Theorem 10.1

A line drawn through the endpoint of a radius and perpendicular to it is a tangent to the circle.

Core Board Corollary: Parallel Tangents at Diameter Ends

The tangents drawn at the ends of any diameter of a circle are **parallel**.

- Let AB be a diameter. Tangent l_1 at A has $\angle OAX = 90^\circ$; tangent l_2 at B has $\angle OBY = 90^\circ$.
- Since interior alternate angles are equal ($\angle OAX = \angle OBY = 90^\circ$), $l_1 \parallel l_2$.
- **Board Rule:** A circle can have at most **two parallel tangents** along any direction.

CHAPTER 10

CORE SYLLABUS

THEOREM 10.2: EQUALITY OF TANGENT LENGTHS

RHS Congruence Proof, Angle Bisectors & Line of Centres

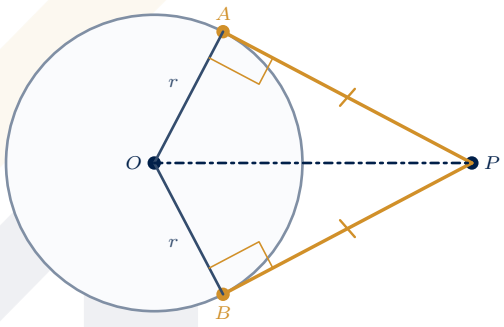
Module 4.1 | Official Theorem Statement

CBSE Theorem 10.2:
The lengths of tangents drawn from an external point to a circle are equal.

$$\boxed{PA = PB}$$

where P is an external point, and A and B are the points of contact of the two tangents.

Module 4.2 | Standard CBSE Board Proof (RHS Congruence Model)



Given: A circle $C(O, r)$ and external point P. Tangents PA and PB touch at A and B.
To Prove: $PA = PB$.
Construction: Join OA, OB, and OP.

Proof Steps:

1. In $\triangle OAP$ and $\triangle OBP$:
2. $\angle OAP = \angle OBP = 90°$
   (Theorem 10.1: Radius $\perp$ Tangent at contact).
3. $OP = OP$ *(Common hypotenuse).*
4. $OA = OB = r$ *(Radii of the same circle).*
5. Therefore, by **RHS Congruence Criterion**:

   $\triangle$OAP $\cong$ $\triangle$OBP

6. By Corresponding Parts of Congruent Triangles (CPCTC):

   PA = PB. **[Hence Proved]**

Module 4.3 | Three Indispensable Corollaries (CPCTC Yields)

Because $\triangle OAP \cong \triangle OBP$, three critical geometric relationships follow immediately:

Corollary 1: **Equal Tangent Angles ($\angle OPA = \angle OPB$):**
OP is the **angle bisector** of the angle between the two tangents ($\angle APB$).

Corollary 2: **Equal Central Angles ($\angle AOP = \angle BOP$):**
OP is the **angle bisector** of the central angle subtended by the contacts ($\angle AOB$).

Corollary 3: **Perpendicular Bisector of Chord ($OP \perp AB$):**
If chord AB connects points of contact, OP is the **perpendicular bisector** of chord AB.

Summary Box: Mirror Symmetry of Line of Centres (OP)

$$PA = PB \qquad \angle OPA = \angle OPB = \frac{1}{2}\angle APB \qquad \angle AOP = \angle BOP = \frac{1}{2}\angle AOB$$

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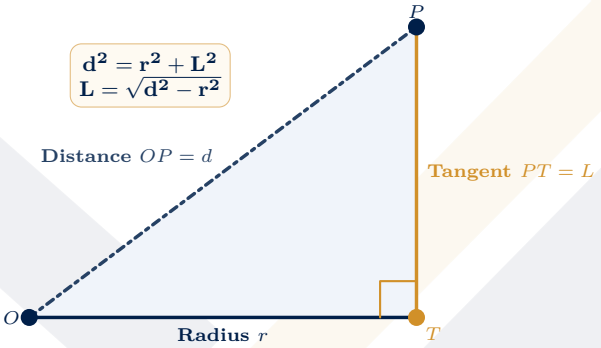
WORKED MODEL

THE TANGENT-RADIUS RIGHT TRIANGLE

Pythagorean Triplet Mechanics & Standard Length Calculations

Module 5.1 | The Fundamental Tangent-Radius Right Triangle

Because Theorem 10.1 guarantees $\angle OTP = 90°$, triangle $\triangle OTP$ is always a **right triangle** with hypotenuse OP:



$$\text{Tangent Length } PT = \sqrt{OP^2 - r^2} \qquad \text{Radius } r = \sqrt{OP^2 - PT^2} \qquad \text{Distance } OP = \sqrt{PT^2 + r^2}$$

Module 5.2 | Standard Pythagorean Triplets in Board Exams

Radius (r)	Tangent (PT)	Distance (OP)	Triplet Family
3 cm	4 cm	5 cm	$(3, 4, 5)$ or scaled $(6, 8, 10)$, $(9, 12, 15)$
5 cm	12 cm	13 cm	$(5, 12, 13)$ or scaled $(10, 24, 26)$
7 cm	24 cm	25 cm	$(7, 24, 25)$
8 cm	15 cm	17 cm	$(8, 15, 17)$

Module 5.3 | Two Core Worked Archetypes

Worked Archetype 1: Calculating Radius from Tangent and Distance

Problem: From point P, the tangent length is 24 cm and distance from centre is 25 cm. Find radius.
Solution: In right $\triangle OTP$ ($\angle OTP = 90°$ by Theorem 10.1):

$$r = \sqrt{OP^2 - PT^2} = \sqrt{25^2 - 24^2} = \sqrt{625 - 576} = \sqrt{49} = \mathbf{7} \text{ cm}.$$

Worked Archetype 2: Calculating Tangent Length

Problem: Point P is 13 cm from centre of a circle of radius 5 cm. Find tangent length.
Solution: In right $\triangle OTP$:

$$PT = \sqrt{OP^2 - OT^2} = \sqrt{13^2 - 5^2} = \sqrt{169 - 25} = \sqrt{144} = \mathbf{12} \text{ cm}.$$

CHAPTER 10

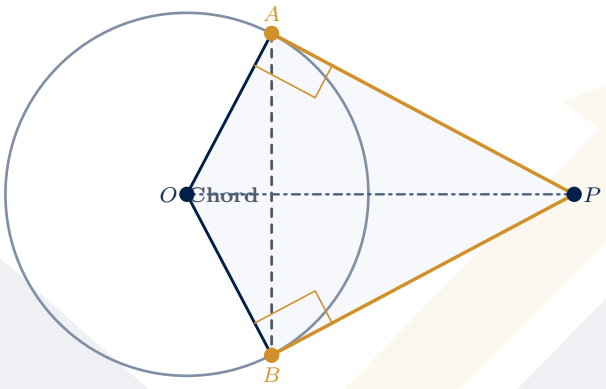
CORE SYLLABUS

ANGLE RELATIONSHIPS & QUADRILATERAL OAPB

Supplementary Angles, Chord Relations & Equilateral Models

Module 6.1 | The Tangent Quadrilateral $OAPB$

When two tangents PA and PB are drawn from external point P, the points form quadrilateral $OAPB$:



The Supplementary Tangent-Angle Theorem:
The angle between two tangents is supplementary to the angle subtended by contact points at centre:

$$\boxed{\angle APB + \angle AOB = 180^\circ}$$

Proof: In quadrilateral $OAPB$: $\angle OAP + \angle AOB + \angle OBP + \angle APB = 360^\circ$.
Since $\angle OAP = \angle OBP = 90^\circ$: $90^\circ + \angle AOB + 90^\circ + \angle APB = 360^\circ \implies \mathbf{\angle AOB + \angle APB = 180^\circ}$.
Deduction: Quadrilateral $OAPB$ is a **cyclic quadrilateral**.

Module 6.2 | Angle with Chord of Contact (AB)

Since $PA = PB$, $\triangle PAB$ is an **isosceles triangle**:

1. $\angle PAB = \angle PBA = \frac{180^\circ - \angle APB}{2} = 90^\circ - \frac{1}{2}\angle APB$.
2. Since $\angle OAP = 90^\circ$, we have $\angle OAB = 90^\circ - \angle PAB = \frac{1}{2}\angle APB$.
3. Hence:

$$\boxed{\angle APB = 2\angle OAB}$$

Module 6.3 | Worked Board Archetype

Worked Archetype 3: Tangents Inclined at 60°

Problem: Two tangents PA, PB have $\angle APB = 60^\circ$. If $OP = 10$ cm, find radius r and tangent length.
Solution:

1. OP bisects $\angle APB$, so $\angle OPA = 30^\circ$. In right $\triangle OAP$ ($\angle OAP = 90^\circ$):
2. $\sin 30^\circ = \frac{r}{OP} \implies \frac{1}{2} = \frac{r}{10} \implies \mathbf{r = 5}$ cm.
3. $\cos 30^\circ = \frac{PA}{OP} \implies \frac{\sqrt{3}}{2} = \frac{PA}{10} \implies \mathbf{PA = 5\sqrt{3}}$ cm.

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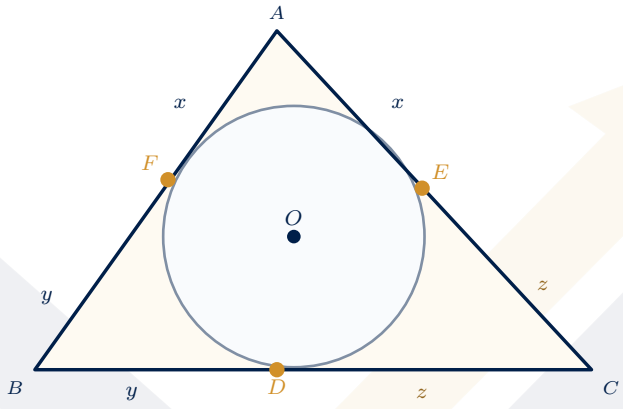
CORE SYLLABUS

TANGENTS IN TRIANGLES & QUADRILATERALS

Incircle Splitting, Circumscribed Quadrilateral ($AB + CD = AD + BC$) & Rhombus

Module 7.1 | Circle Inscribed in a Triangle (Tangent Splitting)

When a circle with centre O and radius r is inscribed in $\triangle ABC$, it touches sides BC, CA, AB at D, E, F:

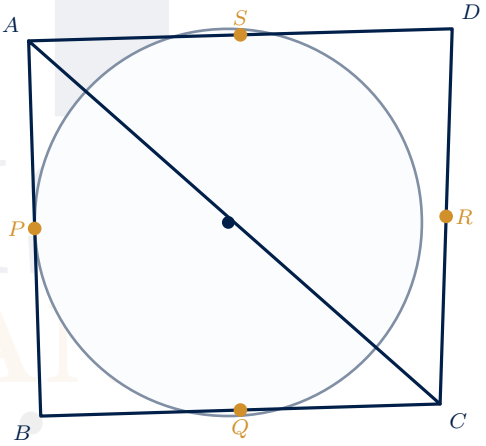


Tangent segments from each vertex are equal: $AF = AE = x$, $BD = BF = y$, $CD = CE = z$.

$$\boxed{AF = x = s - a} \qquad \boxed{BD = y = s - b} \qquad \boxed{CD = z = s - c} \qquad \boxed{\text{Area}(\triangle ABC) = r \cdot s}$$

where $s = \frac{a+b+c}{2}$ is the semi-perimeter.

Module 7.2 | Quadrilateral Circumscribing a Circle (The Pitot Theorem)



CBSE Classic Theorem:
If quadrilateral $ABCD$ circumscribes a circle, then sum of opposite sides are equal:

$$\boxed{AB + CD = AD + BC}$$

Proof: Let the circle touch AB, BC, CD, DA at P, Q, R, S.

$$AP = AS \quad \text{— (1)} \qquad BP = BQ \quad \text{— (2)}$$
$$CR = CQ \quad \text{— (3)} \qquad DR = DS \quad \text{— (4)}$$

Adding (1) + (2) + (3) + (4):

$$(AP + BP) + (CR + DR) = (AS + DS) + (BQ + CQ)$$

$$\mathbf{AB + CD = AD + BC.} \quad \textbf{[Hence Proved]}$$

Circumscribed Parallelogram is a Rhombus

If $ABCD$ is a parallelogram circumscribing a circle: $AB = CD$ and $AD = BC$. Then $AB + CD = AD + BC \implies 2AB = 2AD \implies AB = AD$. Since adjacent sides are equal, $ABCD$ **is a Rhombus**.

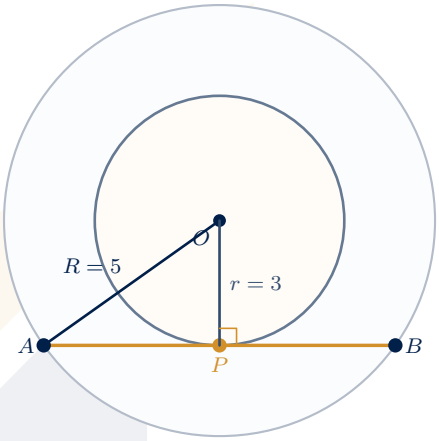
CHAPTER 10

WORKED MODEL

WORKED BOARD-EXAM MASTER MODELS

Concentric Circles, Right-Triangle Incircle & Chord Geometry

Module 8.1 | Model 1: Concentric Circles Chord-Tangent Theorem



Problem (CBSE Standard 3-Mark Question):
Two concentric circles have radii 5 *cm and* 3 *cm. Find the length of the chord of the larger circle which touches the smaller circle.*

Solution:

1. Let O be common centre, C_1 $(R = 5)$ and C_2 $(r = 3)$.
2. Chord AB of C_1 touches C_2 at P.
3. By Theorem 10.1, $OP \perp AB$.
4. Perpendicular from centre to chord bisects the chord: $AP = PB = \frac{1}{2}AB$.
5. In right $\triangle OPA$:

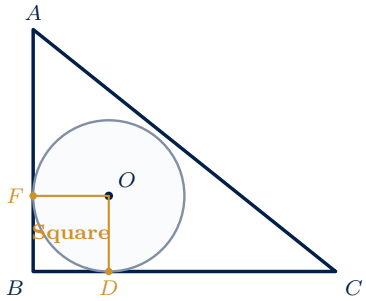
$$AP = \sqrt{R^2 - r^2} = \sqrt{5^2 - 3^2} = \sqrt{16} = 4 \text{ cm}$$

6. Chord $AB = 2 \times AP = 2 \times 4 = \mathbf{8}$ cm.

General Concentric Chord Formula: $\boxed{\text{Length of chord } AB = 2\sqrt{R^2 - r^2}}$

Module 8.2 | Model 2: Incircle of a Right-Angled Triangle

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Problem (High-Frequency Board Archetype):
A circle is inscribed in right $\triangle ABC$ ($\angle B = 90°$). With sides a, c and hypotenuse b, prove inradius:

$$\boxed{r = \frac{a + c - b}{2}}$$

Proof: Let the circle touch BC at D, AB at F, and AC at E.

- Since $OF \perp AB$ and $OD \perp BC$, $OFBD$ has three $90°$ angles and adjacent sides equal ($OF = OD = r$). Thus, $OFBD$ **is a square**.
- Hence, $BF = BD = r$.
- $AE = AF = c - r$ and $CE = CD = a - r$.
- Hypotenuse $b = AE + CE = (c - r) + (a - r) = a + c - 2r$.
- Rearranging: $2r = a + c - b \implies \mathbf{r = \frac{a+c-b}{2}}$.

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CHAPTER 10

EXTENSION PRACTICE

COMMON TANGENTS TO TWO CIRCLES

Relative Positions, Centre Distances & Tangent Counts

Module 9.1 | Direct vs. Transverse Common Tangents

A **common tangent** touches both circles simultaneously:

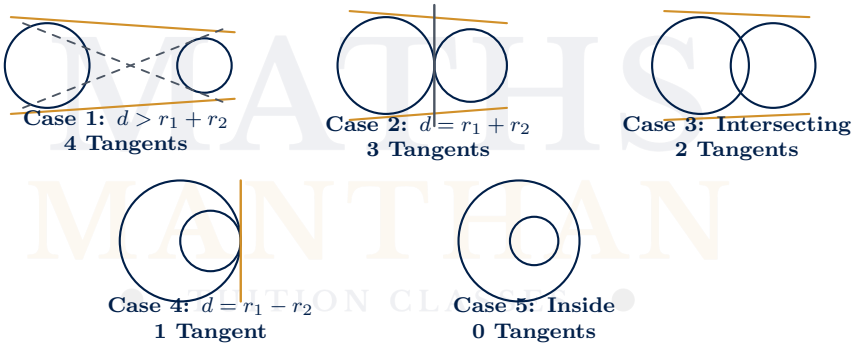
- **Direct (External):** Both circles lie on the **same side** of the tangent line.
- **Transverse (Internal):** Circles lie on **opposite sides** of the tangent (line crosses centre segment).

Module 9.2 | The Five Positional Configurations

Let two circles have radii r_1, r_2 $(r_1 \geq r_2)$ and centre-to-centre distance $d = O_1O_2$:

Case	Configuration	Condition on d	Total	Breakdown
1	**Disjoint (Separate)**	$d > r_1 + r_2$	**4**	2 Direct + 2 Transverse
2	**Touching Externally**	$d = r_1 + r_2$	**3**	2 Direct + 1 Transverse
3	**Intersecting**	$\lvert r_1 - r_2 \rvert < d < r_1 + r_2$	**2**	2 Direct + 0 Transverse
4	**Touching Internally**	$d = r_1 - r_2$	**1**	1 Direct
5	**Concentric / Inside**	$d < r_1 - r_2$	**0**	No common tangent

Module 9.3 | Visual Representations of Common Tangents



Case 1: $d > r_1 + r_2$ **4 Tangents**

Case 2: $d = r_1 + r_2$ **3 Tangents**

Case 3: Intersecting 2 Tangents

Case 4: $d = r_1 - r_2$ **1 Tangent**

Case 5: Inside 0 Tangents

Enrichment / Extension Scope Note

The classification of common tangents is an **extension topic** useful for conceptual clarity and 1-mark objective questions. Length calculations of common tangents are strictly non-evaluative for Class 10 board exams.

CHAPTER 10

REFERENCE MATRIX

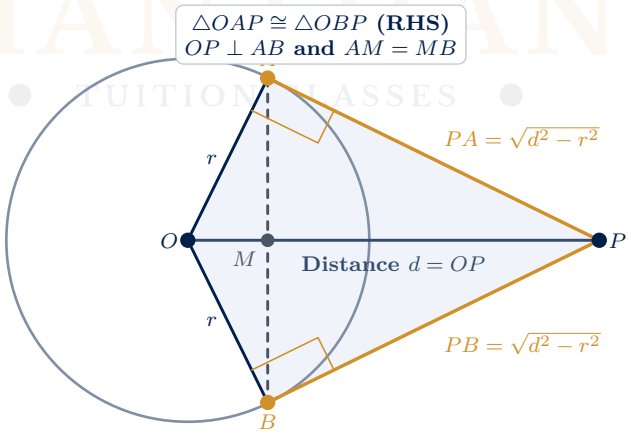
MASTER SUMMARY & FORMULA MATRIX

The Complete Geometric Reference Sheet for Board Quick-Revision

Module 10.1 | Comprehensive Theorem & Formula Repository

Concept / Theorem	Core Formula / Statement	Key Geometric Implication
Theorem 10.1	$OP \perp$ **Tangent at** P	Radius to point of contact makes $90°$. Right triangle for Pythagoras.
Theorem 10.2	$PA = PB$	Tangents from external point are equal in length. Forms isosceles $\triangle PAB$.
OP as Angle Bisector	$\angle OPA = \angle OPB = \frac{1}{2}\angle APB$	Line of centres bisects angle between tangents and central angle.
Supplementary Angles	$\angle AOB + \angle APB = 180°$	Quadrilateral $OAPB$ is cyclic. $\angle$centre $+ \angle$external $= 180°$.
Angle with Chord	$\angle OAB = \frac{1}{2}\angle APB$	Angle between radius and chord is half the angle between tangents.
Pythagoras Relation	$PT = \sqrt{OP^2 - r^2}$	Directly computes tangent length given distance from centre and radius.
Circumscribed Quad	$AB + CD = AD + BC$	Sum of opposite sides of circumscribed quadrilateral are equal.
Circumscribed $\parallel gm$	$ABCD$ **is a Rhombus**	A circumscribed parallelogram has all 4 sides equal.
Concentric Chord	**Chord** $= 2\sqrt{R^2 - r^2}$	Chord of larger circle touching smaller circle is bisected at contact.
Right Triangle Incircle	$r = \frac{a+c-b}{2}$	Rapid formula for inradius given legs a, c and hypotenuse b.
General Triangle Incircle	$\Delta = r \cdot s \implies r = \frac{\Delta}{s}$	Connects triangle area, inradius, and semi-perimeter.

Module 10.2 | Visual Concept Map: Unified Symmetries Blueprint



CHAPTER 10

TRAP ALERT

EXAM TRAPS, HACKS & PRE-SUBMISSION AUDIT

Crucial Board Pitfalls, Instant Verification Checks & Final Protocol

Module 11.1 | Top 8 CBSE Board Exam Traps in Circles

1. **Assuming Hypotenuse is the Tangent:** In $\triangle OTP$, hypotenuse is ALWAYS OP (centre to external point), NOT tangent PT. Write $OP^2 = r^2 + PT^2$, never $PT^2 = OP^2 + r^2$.
2. **Missing Right-Angle Justification in Proofs:** Whenever setting $\angle OAP = 90°$, you MUST cite: *"Radius is perpendicular to the tangent at point of contact (Theorem 10.1)"*.
3. **Incorrect Congruence Criterion for Theorem 10.2:** Do not cite SAS or SSS. The standard NCERT proof strictly uses **RHS** ($\angle OAP = \angle OBP = 90°$, hypotenuse $OP = OP$, radius $OA = OB = r$).
4. **Confusing Central Angle with Arc Angles:** The external tangent angle ($\angle APB$) is supplementary to the central angle ($\angle AOB$), NOT to an inscribed angle on the circumference.
5. **Misordering Segments in Circumscribed Quadrilaterals:** When adding equations $AP = AS, BP = BQ, CR = CQ, DR = DS$, ensure left sides form $AB + CD$ and right sides form $AD + BC$.
6. **Forgetting Chord Bisection by Perpendicular:** In concentric circles, $OP \perp AB$ bisects chord ($AP = PB$). Don't forget the factor of 2: report $AB = 8$ cm, not $AP = 4$ cm.
7. **Unit Inconsistency:** Always convert radius and external distance into the same unit (cm) before applying Pythagoras.
8. **Assuming Circumscribed Quadrilateral is a Rhombus:** A general circumscribed quadrilateral only satisfies $AB + CD = AD + BC$. It is a rhombus ONLY IF already given to be a parallelogram.

Module 11.2 | Arithmetic & Algebraic Verification Hacks

- **The Hypotenuse Benchmark:** Tangent length PT must strictly be **less than** distance OP ($PT < OP$).
- **The** $180°$ **Sanity Check:** Always verify $\angle APB + \angle AOB = 180°$. (e.g. $70° + 110° = 180°$).
- **Right Triangle Inradius Quick Check:** Check $r = \frac{a+c-b}{2}$. For $(6, 8, 10)$: $r = \frac{6+8-10}{2} = 2$ cm.

Module 11.3 | The 5-Point Pre-Submission Board Exam Audit

Check	Verification Item	Board Evaluation Standard
☐ 1	Right-Angle Marks in Diagrams	Did you mark right-angle squares at every point of contact (A, B, T)?
☐ 2	Formal Theorem Citations	Are "Theorem 10.1" and "Theorem 10.2" stated with full geometric reasons?
☐ 3	Congruence State & Reason	Is RHS congruence explicitly stated with 3 criteria (Right angle, Hypotenuse, Radius)?
☐ 4	Doubling Concentric Chords	Did you multiply the half-chord by 2 to give the full chord length AB?
☐ 5	Final Units Explicitly Stated	Are units (cm, m) explicitly written after every numerical answer?

Master the Theorems. Construct the Logic. Secure the 100%.

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